

FinVeritas V2

From Explainable Analysis to Explainable Forecasting — Updated Abstract

Version: 2.1 (incorporates Deterministic Financial Metrics Engine)

Abstract

FinVeritas V1 established a core principle for trustworthy financial AI: Python computes, the LLM narrates. It solved the problem of analyzing historical financial data with credibility and transparency. Version 2 extends this same philosophy from the past into the future — tackling not just "what happened" but "what will happen, why, and how confident should you be."

Before extending into forecasting, V2 first widens the calculation layer itself. A structured feasibility review of Bloomberg's standardized Financial Statement Analysis reports (validated against real filings for TCS, Infosys, and HDFC Bank) confirms that approximately 90% of the quantitative metrics used in professional equity research and corporate financial analysis can be computed directly and deterministically from data FinVeritas already ingests — spanning revenue, cost, profitability, liquidity, solvency, efficiency, return, cash flow, growth, trend, and risk intelligence. This Deterministic Financial Metrics Engine becomes the computational foundation beneath every other V2 capability, closing the gap between raw financial statements and analyst-grade metrics entirely in Python, before any LLM is involved.

The weakest link in any forward-looking financial model (CAGR-based or driver-based) has always been the human-chosen growth assumption. V2 introduces an AI-assisted assumption validation layer that cross-checks a user's growth assumptions against analyst estimates, sector benchmarks, and management guidance extracted directly from earnings calls — flagging unrealistic assumptions before they propagate through the model. Every projected number is paired with a narrated rationale, turning a static forecast into a defensible, challengeable story rather than an opaque figure.

V2 also broadens FinVeritas from single-company analysis into competitive and governance intelligence — benchmarking a company against its market and competitors, and screening executive backgrounds for red flags. A new anomaly alert system monitors key structural ratios (Debt/Equity, Net Asset/Total Debt, etc.) and flags suspicious deviations automatically, reinforcing the platform's core promise: no number reaches the user unverified, and no risk goes unflagged.

V2 also adds a geography-wise sales impact lens: annual reports already disclose segment-wise revenue by region (North America, Europe, APAC, etc.) under standard segment-reporting requirements. FinVeritas parses this disclosed breakdown to compute region-wise revenue share, growth, and volatility in Python, then cross-references regional swings against known regional incidents — currency moves, regional slowdowns, geopolitical disruptions, natural disasters — pulled from the existing news/sentiment pipeline. The LLM narrates why a region over- or under-performed; the underlying regional numbers stay fully deterministic.

An experimental scenario-shock engine (e.g., "what if a recession cuts consumer spending 10% in year 3?") is under evaluation as a possible extension, pending validation of how reliably it can re-run driver models under simulated stress without compromising determinism.

Structured Breakdown of Additions to V1

1. Deterministic Financial Metrics Engine (new — foundational)

- Expands Python-side calculation coverage from a handful of agent-specific ratios to 12 categories of financial intelligence: Revenue, Cost, Profitability, Liquidity, Solvency, Debt Servicing, Efficiency, Return, Cash Flow, Growth, Trend, and Risk metrics

- Feasibility validated against real Bloomberg filings (TCS, Infosys, HDFC Bank) — ~90% of metrics computable purely from data already ingested, with zero LLM computation involved
- The remaining ~10% (market valuation multiples, precise DSCR, governance scores, competitive benchmarks) requires external data feeds — this gap directly motivates and feeds the Competitive Intelligence Agent and Executive Background Check Agent described below
- Every metric is pre-computed and locked before being handed to any agent for narration, extending the “Python computes, LLM narrates” guarantee across roughly 10x more financial indicators than V1

2. Assumption Validation Agent (new)

- Searches for analyst estimates, industry growth forecasts, competitor performance
- Parses earnings call transcripts / management guidance to extract expected growth
- Flags assumptions that deviate significantly from sector norms
- Still Python-anchored: the validation logic and comparison math is deterministic; the LLM only explains the discrepancy

3. Reasoning Layer (“Why” Narration) (new)

- Moves beyond reporting a number to narrating causal logic (e.g., decelerating ARPU growth → slower revenue growth, benchmarked against comparable companies)
- Converts every forecast into a stakeholder-reviewable argument, not just an output

4. Competitive Intelligence Agent (new)

- Analyzes the broader market and positions the company against direct competitors
- Extends the existing multi-source data ingestion pipeline (Bloomberg, Yahoo Finance, uploads) to include competitor data
- Consumes the market-data and peer-benchmark gap identified by the Metrics Engine feasibility review

5. Executive Background Check Agent (new)

- Screens leadership history for red flags (legal issues, prior company failures, conflicts of interest)
- Adds a governance/trust dimension alongside financial trust

6. Anomaly Alert System (new)

- Continuously monitors key structural financial ratios — Debt/Equity, Net Asset/Total Debt, and similar leverage/solvency indicators
- Triggers alerts when values breach safe thresholds or diverge from historical/sector patterns
- Extends the Credibility Engine's PASS/WARN/FAIL logic into an ongoing risk-monitoring layer rather than a one-time data check

7. Working Capital & Credit Intelligence Agent (new)

- Extends FinVeritas beyond financial statement analysis into banking and credit intelligence by evaluating working capital requirements, computing Maximum Permissible Bank Finance (MPBF), liquidity position, and debt servicing capability. This enables explainable preliminary credit appraisal for listed companies while maintaining the platform's core philosophy: Python computes, LLM narrates.

8. Geography-wise Impact Analysis Agent (new)

- Extracts geography/segment-wise revenue disclosures already present in annual reports (region-wise sales breakdown under segment reporting standards) — no new data source required, only new parsing logic
- Computes region-wise revenue share, YoY growth, and volatility deterministically in Python, the same way the Metrics Engine computes company-level figures
- Cross-references regional performance swings against known regional incidents (currency movements, regional slowdowns, geopolitical disruptions, natural disasters) sourced via the existing Sentiment Agent's news pipeline

- LLM narrates the likely cause of regional over-/under-performance; the underlying regional revenue figures remain Python-computed and unverified narration never overrides them

Appendix: Metrics Engine Feasibility Summary

Coverage achievable directly from Bloomberg-standardized financial statements, validated against real company filings:

Metric Category	Feasibility
Income Statement Metrics	100%
Balance Sheet Metrics	100%
Cash Flow Metrics	100%
Profitability Ratios	100%
Liquidity Ratios	100%
Solvency Ratios	100%
Return Ratios	100%
Efficiency Ratios	~95%
Growth Metrics	100%
Trend Analysis	100%
Risk Monitoring	~95%
Geographic Segment Sales (region-wise revenue)	~85% (depends on disclosure granularity)
Market Valuation (P/E, EV/EBITDA, PEG)	Requires market data
Precise DSCR	Requires debt repayment schedule
Governance Intelligence	Requires external data
Competitive Benchmarking	Requires external data

Conclusion: The Bloomberg Financial Statement Analysis reports provide a sufficiently rich, standardized dataset for FinVeritas V2 to evolve into a comprehensive Financial Metrics Intelligence Engine. Roughly 90% of professional-grade financial metrics can be computed deterministically today; the remaining advanced capabilities (market valuation, precise debt servicing, governance, and competitive benchmarking) are unlocked as V2's new agents integrate external data feeds.